15.095: Machine Learning Under a Modern Optimization Lens **Fall 2026**

Homework 4: Due October 28

Hand in: **pdf** upload to Gradescope. Please append any Julia code **at the end** of the whole pdf.

Note: this homework covers lectures 11–14. We recommend attempting questions after the relevant content has been covered in class. Total: 100 points.

Question 1: True/False (30 points)

Please classify the following statements as true or false and justify your answer. We will assign 1 point for correctly classifying the answer, and 5 points for the validity of the justification/counterexample.

(a) (6 points) Let $\mathcal{Z} = \mathbb{R}$ and $\lambda > 0$. With the linear kernel $K(\mathbf{x}, \mathbf{x}') = \mathbf{x}^\top \mathbf{x}'$, the kernel regression prediction of the optimizer gives the same decision rules as a linear rule $\hat{z}(\mathbf{x}) = \boldsymbol{\beta}^\top \mathbf{x}$ whose $\boldsymbol{\beta}$ minimizes the average training cost $\frac{1}{n}\sum_{i=1}^{n} c(\boldsymbol{\beta}^\top \mathbf{x}^i; y^i)$ plus the ridge penalty $\lambda\|\boldsymbol{\beta}\|_2^2$.

(b) (6 points) The optimal design of experiments formulation assigns $n = mk$ subjects to m groups of k and includes the constraints $x_{ip} = 0$ for all $i < p$. These constraints can cut off every optimal assignment, so they speed up the solver at the price of a possibly larger discrepancy.

(c) (6 points) Because the optimal design of experiments approach forms the groups deterministically rather than at random, the difference in mean outcomes between two treatment groups is a biased estimate of the difference in their treatment effects.

(d) (6 points) In ROAD I, prognostic matching leaves equally many treated and untreated patients in every risk bucket, so a patient's bucket carries no information about whether they received imatinib. The matched cohort is therefore free of confounding, as in a randomized trial.

(e) (6 points) ROAD II balances a latent factor $\tilde{U}$, which summarizes each patient's unobserved prognostic factors, between treated and untreated patients. On data from a randomized trial, this balancing should leave the estimated treatment effect nearly unchanged.

Question 2: Two Rewards, Two Policies (20 points)

A hospital treats a chronic condition with one of two drugs, A and B. Its protocol gives drug B to a severe patient with probability 0.8 and to a mild patient with probability 0.2. You want to learn, from the hospital's records, a policy that prescribes the drug maximizing survival.

For patient i you observe $x_i = 1$ if the disease is severe and $x_i = 0$ if it is mild, the drug $T_i \in \{\mathrm{A}, \mathrm{B}\}$ the patient received, and the survival y_i, in years. Models f_{A} and f_{B} predict survival under each drug, and p_{it} is the probability that patient i receives drug t. Four patients are below.

i	x_i	T_i	y_i	$f_{\mathrm{A}}(x_i)$	$f_{\mathrm{B}}(x_i)$	$p_{i\mathrm{A}}$	$p_{i\mathrm{B}}$
1	0	A	6	6	7	0.8	0.2
2	0	B	4	6	7	0.8	0.2
3	1	B	5	3	5	0.2	0.8
4	1	A	2	3	5	0.2	0.8

Recall from lecture the direct method (DM) and doubly robust (DR) rewards,

$$\Gamma^{\mathrm{DM}}_{it} = f_t(x_i), \qquad \Gamma^{\mathrm{DR}}_{it} = f_t(x_i) + \mathbf{1}\{T_i = t\}\,\frac{y_i - f_t(x_i)}{p_{it}}.$$

A policy π is scored by its total reward $\sum_{i=1}^{4} \Gamma_{i\pi(x_i)}$. A *policy tree of depth at most* 1 either gives one drug to every patient, or splits on x and gives one drug to the mild leaf and one to the severe leaf.

(a) (7 points) Compute the 4×2 reward matrix Γ^{DR}.

(b) (8 points) For Γ^{DM} and for Γ^{DR}, find the policy tree of depth at most 1 with the largest total reward. Report the drug it gives to mild and to severe patients, and its total reward.

(c) (5 points) The two methods disagree on mild patients. Explain why, naming the patient responsible, and say which policy you would trust.

Final Project: Project Proposal (50 points)

The final team project is 30% of your grade. Its goal is to use a method from this course to solve a real-world problem, or a well-motivated theoretical one. For the last part of this homework assignment, you and your selected team will be expected to submit a project proposal. We expect **one submission per team on GradeScope, please add teammates into submission**. There will be a **different submission** slot for the proposal, separate from the individual homework.

Teams. A team has two students. You may work alone or in a team of three after talking with the instructor.

Topic. The topic is your choice. A good project has a clear question, data or a model to work with, and a method from the course at its center. The method can be a formulation you solve with mixed-integer optimization, a tree-based learner, a prescriptive or robust method, or an extension of one of them, applied to a problem you care about. Examples of the kinds of problem that fit:

- applying some regularized or constrained regression for some real-world problem;
- learning an interpretable rule from observational data, with optimal classification, regression or policy trees;
- deciding what to do from data, with the prescriptive methods of Lectures 10 and 11;
- filling in missing entries, grouping observations or designing an experiment, with the methods of the later lectures;
- a (well-motivated) theoretical question about one of these methods, with an appropriate scope given the duration of the class

There will be real examples for past years posted to Canvas in the following days.

Deadlines.

Proposal	October 28, with this problem set
Presentation	December 8
Final report (8 pages)	December 14

Proposal. Submit one proposal per team. The proposal should be written as a LaTeX or Word/Google Docs, and submitted as a PDF to GradeScope. For formatting, please use ≤ 3 pages (excluding references), with rough breakdown of: 1/2 to 1 page of motivation, 1/2 to 1 page of methods, 1/2 to 1 page of evaluation information. It should have the following components:

- (20 points) **Motivation**: introduce the problem, relevant background information, and your proposed plan of action. Please include relevant references.
- (20 points) **Proposed Methods**: explain the course method you plan to apply, why this method is appropriate, and document any potential challenges given the problem.
- (10 points) **Evaluation** explain how you will check that it worked; for example against a baseline (please then enumerate baseline methods, and reason for using these)